# Math 107 Midterm 2 practice

Last updated: November 5, 2026.

Based on Lectures 18–25 and Quizzes 5–9. Use with the study guide. Exam: Friday, November 13 · in class (not posted) · day page.

Three types of problems will appear on Midterm 2: short answer definitions and meaning, hand computations, and short word problems. No software syntax.

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## A. Definitions and meaning

Write a short answer (one or two sentences, or a formula).

**A1.** What does it mean to say that $\mathbf{A}$ is an $m \times n$ matrix? What size is each column of $\mathbf{A}$?

**A2.** Notation: what does the entry $A_{ij}$ mean? What does the column vector $\mathbf{a}_j$ mean? For

$$\mathbf{A} = \begin{bmatrix} 3 & -1 & 4 \\ 0 & 2 & 5 \end{bmatrix},$$

find $A_{12}$ and write $\mathbf{a}_2$.

**A3.** Give the **column view** formula for $\mathbf{Ax}$ when $\mathbf{A}$ has columns $\mathbf{a}_1, \dots, \mathbf{a}_n$ and $\mathbf{x} = (x_1, \dots, x_n)$.

**A4.** In $\mathbf{y} = \mathbf{Ax}$, what are the **input**, **output**, and **transform**? If $\mathbf{y}$ has size $m$ and $\mathbf{x}$ has size $n$, what is the size of $\mathbf{A}$?

**A5.** The structured types from class (beyond "square") are: **zero**, **identity**, **diagonal**, **triangular**, **symmetric**, and **sparse**. For each matrix, circle **all** types that apply.

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & -1 \\ 0 & 0 & 4 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 0 & 0 & 5 \\ 0 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix}.$$

**A6.** Translate "$\mathbf{Ax} = \mathbf{b}$ has a solution" into a statement about the **column space** of $\mathbf{A}$.

**A7.** Fill in the blanks for the RREF-by-hand outline (assume each candidate pivot you meet is nonzero — no row swap needed).

Start at the _____ corner; call that entry a _____. Scale its row so the pivot equals ____. Eliminate _____ the pivot (zeros in that column). Move to the next _____ entry; eliminate _____ and _____ that pivot. Continue until the _____ corner (or you run out of rows/columns).

**A8.** Each matrix below is presented as the RREF of an augmented matrix $\mathbf{M} = [\mathbf{A} \,|\, \mathbf{b}]$. For each, say whether it is actually in **RREF** (yes/no). If no, give a brief reason.

$$\mathbf{R}_1 = \left[\begin{array}{ccc|c} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right], \quad \mathbf{R}_2 = \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{array}\right], \quad \mathbf{R}_3 = \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{array}\right].$$

**A9.** The RREF of the augmented matrix $\mathbf{M} = [\mathbf{A} \,|\, \mathbf{b}]$ is

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 4 \\ 0 & 1 & 3 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

List the **pivot** variables and the **free** variables (use $x_1, x_2, x_3$).

**A10.** Match each solution type for $\mathbf{Ax} = \mathbf{b}$ with the corresponding RREF description. Write the three pairs (e.g. 1–b).

| | Solution type | | RREF description |
|---|---|---|---|
| 1 | no solution | a | consistent, and at least one free variable |
| 2 | unique solution | b | inconsistent row $[0 \ \cdots \ 0 \mid c]$ with $c \neq 0$ |
| 3 | infinitely many solutions | c | consistent, and no free variables |

**A11.** Write the model for a **linear dynamical system** used in class. What do the state $\mathbf{x}_t$ and matrix $\mathbf{A}$ represent?

**A12.** When is the product $\mathbf{AB}$ defined? What are the dimensions of $\mathbf{AB}$ if $\mathbf{A}$ is $m \times n$ and $\mathbf{B}$ is $n \times p$?

**A13.** Does the order generally matter in matrix multiplication, e.g. does $AB = BA$ for square $\mathbf{A}$ and $\mathbf{B}$?

**A14.** How does $\mathbf{A}^k\mathbf{x}$ relate to the dynamical system $\mathbf{x}_{t+1} = \mathbf{A}x_t$?

**A15.** What does it mean for a square matrix $\mathbf{A}$ to be **invertible**? How does invertibility relate to uniqueness of solutions of $\mathbf{Ax} = \mathbf{b}$?

**A16.** Define the **column space** $C(\mathbf{A})$ and the **null space** $N(\mathbf{A})$.

**A17.** State the **rank–nullity** theorem for an $m \times n$ matrix $\mathbf{A}$ (e.g. $n = \operatorname{rank}(\mathbf{A}) + \cdots$).

**A18.** For a $2 \times 2$ matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, write $\det \mathbf{A}$. What does $|\det \mathbf{A}|$ mean geometrically for the image of the unit square?

**A19.** True or false, with a one-line reason: "If $\det \mathbf{A} = 0$, then $\mathbf{A}$ is invertible."

**A20.** Light review (Quiz 5): What is a **basis** of a subspace $\mathbf{V}$? What is $\dim \mathbf{V}$?

**A21.** The $2 \times 2$ elementary-matrix templates from class (built by doing the row operation to $\mathbf{I}$) are:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \ \text{(swap)}, \qquad \begin{bmatrix} c & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & c \end{bmatrix} \ \text{(scale row by } c \neq 0\text{)},$$

$$\begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix}, \begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix} \ \text{(add } c \text{ times one row to the other)}.$$

Write the elementary matrix for each row operation: (a) swap the two rows; (b) multiply row 2 by $-3$; (c) add 2 times row 1 to row 2.

**A22.** For each elementary matrix, state the corresponding row operation on a 2-row matrix.

$$\mathbf{E}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}, \quad \mathbf{E}_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{E}_3 = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}.$$

**A23.** Write the $2 \times 2$ inverse formula for $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ (when it exists). What goes wrong if $\det \mathbf{A} = 0$?

**A24.** For each expression, say whether the result is a **scalar**, an $n$-vector, an $m \times n$ matrix, a **set of vectors**, or **undefined** (brief reason if undefined). Assume sizes make sense unless noted.

(a) $\mathbf{Ax}$ when $\mathbf{A}$ is $m \times n$ and $\mathbf{x}$ is an $n$-vector

(b) $\mathbf{A} + \mathbf{B}$ when $\mathbf{A}$ and $\mathbf{B}$ are both $m \times n$

(c) $\mathbf{AB}$ when $\mathbf{A}$ is $2 \times 3$ and $\mathbf{B}$ is $2 \times 3$

(d) $\mathbf{A}^\mathsf{T}$ when $\mathbf{A}$ is $m \times n$

(e) $\det \mathbf{A}$ when $\mathbf{A}$ is $2 \times 2$

(f) $\mathbf{A}^{-1}$ when $\mathbf{A}$ is singular $2 \times 2$

(g) $C(\mathbf{A})$

(h) $N(\mathbf{A})$

(i) $\operatorname{rank}(\mathbf{A})$

(j) $\mathbf{A}^2$ when $\mathbf{A}$ is $3 \times 2$

(k) $\mathbf{A}Bx$ when $\mathbf{A}$ is $2 \times 2$, $\mathbf{B}$ is $2 \times 2$, and $\mathbf{x}$ is a 2-vector

(l) $\mathbf{A}\mathbf{x} - \mathbf{b}$ when $\mathbf{A}\mathbf{x} = \mathbf{b}$ is consistent

(m) $\mathbf{E}\mathbf{A}$ when $\mathbf{E}$ is $2 \times 2$ elementary and $\mathbf{A}$ is $2 \times 3$

(n) $\dim N(\mathbf{A})$ when $\mathbf{A}$ is $3 \times 5$ of rank 2

**A25.** True or false: "If $N(\mathbf{A}) = \{0\}$, then $\mathbf{A}\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b}$."

**A26.** Let $\mathbf{A}$ be square $n \times n$. Match each condition to whether it means $\mathbf{A}$ is **invertible** or **not invertible**. Write pairs (e.g. 1–a).

| | Condition | | Conclusion |
|---|---|---|---|
| 1 | $\det \mathbf{A} \neq 0$ | a | invertible |
| 2 | $\operatorname{rank}(\mathbf{A}) < n$ | b | not invertible |
| 3 | $N(\mathbf{A}) = \{0\}$ | | |
| 4 | $\mathbf{A}\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b}$ | | |
| 5 | RREF of $\mathbf{A}$ is $\mathbf{I}$ | | |

**A27.** In the dynamical model $\mathbf{x}_{t+1} = \mathbf{A}x_t$, what does the $(i, j)$ entry of $\mathbf{A}$ mean if columns are "from" and rows are "to"?

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# B. Computational

Show your work for each calculation.

## Matrices, Ax, and transforms

**B1.** Let

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}.$$

Compute $\mathbf{A}\mathbf{x}$ using the column-combination view.

**B2.** Let

$$\mathbf{A} = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

Compute $\mathbf{A}\mathbf{x}$ by hand (row–column or column combination).

**B3.** Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}.$$

Compute $\mathbf{A}\mathbf{x}$ by hand.

**B4.** Let

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{bmatrix}.$$

In $\mathbf{y} = \mathbf{A}\mathbf{x}$, what are the sizes of the input $\mathbf{x}$ and the output $\mathbf{y}$?

**B5.** Let

$$\mathbf{A} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{x}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Compute $\mathbf{y}_1 = \mathbf{A}\mathbf{x}_1$ and $\mathbf{y}_2 = \mathbf{A}\mathbf{x}_2$. Sketch each $\mathbf{x}_i$ and its image $\mathbf{y}_i$ in the plane. Then describe the geometric effect of $\mathbf{y} = \mathbf{A}\mathbf{x}$.

**B6.** Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{x}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Compute $\mathbf{y}_1 = \mathbf{A}\mathbf{x}_1$ and $\mathbf{y}_2 = \mathbf{A}\mathbf{x}_2$. Sketch each $\mathbf{x}_i$ and its image $\mathbf{y}_i$ in the plane. Then say what kind of transform $\mathbf{y} = \mathbf{A}\mathbf{x}$ is (scaling / reflection / shear / rotation).

**B7.** Let

$$\mathbf{A} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Compute $\mathbf{A} + \mathbf{B}$, $2\mathbf{A}$, and $\mathbf{A}^{\mathsf{T}}$.

**B8.** Is

$$\mathbf{T} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

symmetric? Triangular? Diagonal?

**B9.** Let

$$\mathbf{F} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 2 & 0 \\ 0 & 0.5 \end{bmatrix}.$$

Compute $\mathbf{D}\mathbf{F}\begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

**B10.** Let $\mathbf{A} = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 0 \end{bmatrix}$. Write $\mathbf{A}^{\mathsf{T}}$ and give its size.

**B11.** Let

$$\mathbf{R} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{P} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Compute $\mathbf{R}\mathbf{P}$.

## Systems, elimination, RREF

**B12.** Write the linear system

$$\begin{aligned} x + 2y &= 5, \\ 3x - y &= 1 \end{aligned}$$

as $\mathbf{A}\mathbf{x} = \mathbf{b}$. Give $\mathbf{A}$, $\mathbf{x}$, and $\mathbf{b}$.

**B13.** Solve by hand (elimination or substitution):

$$\begin{aligned} x + 2y &= 5, \\ 3x - y &= 1. \end{aligned}$$

How many solutions?

**B14.** The augmented matrix $[\mathbf{A} \mid \mathbf{b}]$ row-reduces to

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Identify pivot and free variables. Write the general solution.

**B15.** The augmented matrix $[\mathbf{A} \mid \mathbf{b}]$ row-reduces to

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 4 \end{array}\right].$$

How many solutions?

**B16.** Reduce

$$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 2 & 4 & 6 \end{array}\right]$$

to RREF by hand. How many solutions? Parametrize if infinite.

**B17.** Let $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ and $\mathbf{b} = (3, 7)$. Is $\mathbf{b}$ in $C(\mathbf{A})$? Brief reason.

**B18.** *RREF worksheet.* An augmented matrix has already been reduced to

$$\left[\begin{array}{cccc|c} 1 & 3 & 0 & -1 & 2 \\ 0 & 0 & 1 & 4 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Fill in:

| Item | Answer |
|---|---|
| Pivot variables | |
| Free variables | |
| Rank of $\mathbf{A}$ | |
| Consistent? | |
| General solution $\mathbf{x} =$ | |

**B19.** The elementary matrix for "add $-2$ times row 1 to row 2" on a $2 \times n$ matrix is $\mathbf{E} = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$. Compute $\mathbf{E}\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 2 & 4 & 6 \end{array}\right]$ and read the solution set.

## Dynamics

**B20.** Let

$$\mathbf{A} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}, \quad \mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Compute $\mathbf{x}_1 = \mathbf{A}x_0$ and $\mathbf{x}_2 = \mathbf{A}x_1$.

**B21.** A region's population is split into urban and rural. Let $\mathbf{x}_t = (u_t, r_t)$ be the fractions urban and rural in year $t$. Each year: 90% of urban residents stay urban and 10% move rural; 80% of rural residents stay rural and 20% move urban. The transition matrix (columns = "from") is

$$\mathbf{A} = \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}.$$

If $\mathbf{x}_0 = (0.6, 0.4)$, compute $\mathbf{x}_1 = \mathbf{A}\mathbf{x}_0$ and interpret the two entries.

**B22.** Let

$$\mathbf{A} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}.$$

For the system $\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t$, express $\mathbf{x}_3$ in terms of $\mathbf{A}$ and $\mathbf{x}_0$ using a matrix power (do not expand unless you want to check).

**B23.** Two rooms: each day, 70% in Room A stay and 30% move to B; 60% in B stay and 40% move to A. Let $\mathbf{x}_t = (a, \mathbf{b})$.

(a) Write $\mathbf{A}$ so $\mathbf{x}_{t+1} = \mathbf{A}x_t$ (columns = from).
(b) If $\mathbf{x}_0 = (50, 50)$, compute $\mathbf{x}_1$.
(c) Does total occupancy stay 100?

## Matrix products and powers

**B24.** Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Compute $\mathbf{AB}$ and $\mathbf{BA}$. Are they equal?

**B25.** Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

Compute $\mathbf{A}^2$.

**B26.** Let

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Compute $\mathbf{A}^2$ and $\mathbf{A}^3$. Guess a formula for $\mathbf{A}^k$.

**B27.** Let $\mathbf{A} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$. Compute $\mathbf{A}^2$ and $\mathbf{A}^3$. Guess $\mathbf{A}^k$.

**B28.** Let $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. Compute $\mathbf{A}^2$ and $\mathbf{A}^3\mathbf{x}$ for $\mathbf{x} = (1, 1)$. Relate to two steps of $\mathbf{x}_{t+1} = \mathbf{A}x_t$.

## Inverses

**B29.** Find $\mathbf{A}^{-1}$ (if it exists) by reducing $[\mathbf{A} \mid \mathbf{I}]$ to $[\mathbf{I} \mid \mathbf{A}^{-1}]$:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

**B30.** Is

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

invertible? Give two different reasons (determinant or null space).

**B31.** Solve

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

without forming an inverse explicitly.

**B32.** Use the $2 \times 2$ formula to invert $\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$. Check $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$.

**B33.** Suppose reducing $[\mathbf{A} \mid \mathbf{I}]$ for a $2 \times 2$ matrix yields

$$\left[\begin{array}{cc|cc} 1 & 0 & 3 & -1 \\ 0 & 1 & -2 & 1 \end{array}\right].$$

What is $\mathbf{A}^{-1}$? What was $\det \mathbf{A}$ if $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$?

## Rank, column space, null space

**B34.** Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 1 \end{bmatrix}.$$

Describe $C(\mathbf{A})$. Is $\mathbf{b} = (1, 2, 3)$ in $C(\mathbf{A})$? Is $\mathbf{b} = (1, 3, 0)$ in $C(\mathbf{A})$?

**B35.** Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 1 \end{bmatrix}.$$

Find $N(\mathbf{A})$. Give a basis for $N(\mathbf{A})$, or state that only 0 is in the null space.

**B36.** The RREF of a $3 \times 4$ matrix $\mathbf{A}$ is

$$\begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

What is $\operatorname{rank}(\mathbf{A})$? Which original columns form a basis for $C(\mathbf{A})$ (pivot columns)? Find a basis for $N(\mathbf{A})$.

**B37.** The RREF of a $3 \times 4$ matrix $\mathbf{A}$ is

$$\begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

What is $\dim N(\mathbf{A})$? Check $\operatorname{rank}(\mathbf{A}) + \dim N(\mathbf{A}) = n$.

**B38.** Let $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$.

(a) Find $\operatorname{rank}(\mathbf{A})$.
(b) Find a basis for $C(\mathbf{A})$ and for $N(\mathbf{A})$.
(c) Is $\mathbf{b} = (1, 2)$ in $C(\mathbf{A})$? Is $\mathbf{b} = (1, 3)$?

## Determinants

**B39.** Compute

$$\det \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}.$$

**B40.** Compute

$$\det \begin{bmatrix} 2 & 1 & 0 \\ 0 & -3 & 4 \\ 0 & 0 & 5 \end{bmatrix}.$$

**B41.** The map $\mathbf{A} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ sends the unit square to a rectangle. What is the area of that rectangle? How does this relate to $\det \mathbf{A}$?

**B42.** Solve by elimination:

$$\begin{aligned} x - y &= 1, \\ 2x + y &= 5. \end{aligned}$$

How many solutions?

**B43.** The RREF of $[\mathbf{A} \mid \mathbf{b}]$ is

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 4 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Write the general solution. What is rank($\mathbf{A}$)?

**B44.** Let $\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$.

(a) Find a basis for $C(\mathbf{A})$ and state rank($\mathbf{A}$).
(b) Find a basis for $N(\mathbf{A})$.
(c) Check rank–nullity.

**B45.** Compute $\det \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$ and, if possible, $\mathbf{A}^{-1}$ by the $2 \times 2$ formula.

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## C. Word problems

Show your work. Write equations and matrices carefully, then interpret the final answer in context.

**C1.** A campus coffee cart sells only lattes and drip coffee. On Monday the cart sold **120** drinks and took in **\$430**. Lattes cost **\$4** each and drip coffee costs **\$3** each. Let $x$ be the number of lattes sold and $y$ the number of drip coffees sold.

(a) Write a system of two linear equations for $x$ and $y$.
(b) Write the system as an augmented matrix $[\mathbf{A} \mid \mathbf{b}]$.
(c) Use elementary row operations (no row swaps needed) to put the augmented matrix into **echelon form**.
(d) Solve for $x$ and $y$.
(e) State the answer in context (how many of each drink were sold).

**C2.** A food truck makes two burrito fillings. Each order of filling A uses **2** cups of beans and **1** cup of rice. Each order of filling B uses **1** cup of beans and **3** cups of rice. Today the truck has **16** cups of beans and **13** cups of rice to use completely. Let $x_1$ and $x_2$ be the number of orders of A and B.

(a) Write the system as $\mathbf{Ax} = \mathbf{b}$ (give $\mathbf{A}$, $\mathbf{x}$, and $\mathbf{b}$).
(b) Using elementary matrices / row operations, clear below the first pivot to reach an upper-triangular $\mathbf{U}$. Write the elementary matrix $\mathbf{E}$ for that step and check $\mathbf{EA} = \mathbf{U}$.
(c) Invert $\mathbf{E}$ to get $\mathbf{L} = \mathbf{E}^{-1}$, and verify $\mathbf{A} = \mathbf{LU}$.
(d) Solve $\mathbf{Ly} = \mathbf{b}$ by forward substitution, then $\mathbf{Ux} = \mathbf{y}$ by back substitution.
(e) State how many orders of each filling the truck should make.

**C3.** Two dorm buildings, North and South, share a shuttle. Each week: **80%** of North residents stay North and **20%** move South; **70%** of South residents stay South and **30%** move North. Let $\mathbf{x}_t = (n_t, s_t)$ be the number of residents in North and South at the start of week $t$.

(a) Write the transition matrix $\mathbf{A}$ so that $\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t$ (columns = "from").
(b) If $\mathbf{x}_0 = (200, 100)$, compute $\mathbf{x}_1 = \mathbf{A}\mathbf{x}_0$.
(c) Does the total number of residents stay 300? Briefly why, in terms of $\mathbf{A}$ or the story.
(d) Express $\mathbf{x}_2$ using a matrix power (you need not multiply it out).

**C4.** A workshop runs two machines. If $\mathbf{x} = (x_1, x_2)$ is hours on machine 1 and machine 2, the output is

$$\mathbf{y} = \mathbf{Ax}, \qquad \mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix},$$

where $\mathbf{y} = (\text{parts}, \text{scrap})$.

(a) From $\mathbf{A}$, find a basis for the column space $C(\mathbf{A})$ and a basis for the null space $N(\mathbf{A})$.
(b) Sketch $C(\mathbf{A})$ and $N(\mathbf{A})$ in $\mathbb{R}^2$ (both are subspaces through the origin — say what geometric object each is).
(c) State rank($\mathbf{A}$) and $\dim N(\mathbf{A})$, and check that rank–nullity holds for this $2 \times 2$ matrix.
(d) In one sentence: what does (b)+(c) say about which output vectors $\mathbf{y}$ are possible, and about uniqueness of $\mathbf{x}$ when a solution exists?

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# Answers

Check only after a closed-book attempt. Update your definitions/formulas list for every miss.

## A — Definitions (key points)

**A1.** $\mathbf{A}$ has $m$ rows and $n$ columns; each column is an $m$-vector.

**A2.** $A_{ij}$ is the entry in row $i$, column $j$; $\mathbf{a}_j$ is the $j$th column of $\mathbf{A}$. Here $A_{12} = -1$ and $\mathbf{a}_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$.

**A3.** $\mathbf{Ax} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$.

**A4.** Input $\mathbf{x}$, output $\mathbf{y}$, transform $\mathbf{A}$. Size of $\mathbf{A}$ is $m \times n$.

**A5.** $\mathbf{A}$: triangular. $\mathbf{B}$: identity, diagonal, triangular, symmetric. $\mathbf{C}$: sparse.

**A6.** $\mathbf{b} \in C(\mathbf{A})$ (equivalently: $\mathbf{b}$ is a linear combination of the columns).

**A7.** Upper-left; pivot; 1; below; diagonal; above; below; lower-right.

**A8.** $\mathbf{R}_1$: yes. $\mathbf{R}_2$: no — not reduced (nonzero entry above the pivot in column 2). $\mathbf{R}_3$: no — pivot in row 2 is 2, not 1.

**A9.** Pivot: $x_1, x_2$. Free: $x_3$.

**A10.** 1–b, 2–c, 3–a.

**A11.** $\mathbf{x}_{t+1} = \mathbf{A}x_t$; $\mathbf{x}_t$ is the state, $\mathbf{A}$ is the transition / update matrix.

**A12.** Inner dimensions match ($\mathbf{A}$'s columns = $\mathbf{B}$'s rows); $\mathbf{AB}$ is $m \times p$.

**A13.** Order matters: $\mathbf{AB}$ and $\mathbf{BA}$ need not both be defined, and even when both are, they need not be equal.

**A14.** $\mathbf{x}_k = \mathbf{A}^k\mathbf{x}_0$ (after $k$ steps).

**A15.** There exists $\mathbf{A}^{-1}$ with $\mathbf{AA}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$. For square $\mathbf{A}$, invertible iff $\mathbf{Ax} = \mathbf{b}$ has a unique solution for every $\mathbf{b}$.

**A16.** $C(\mathbf{A}) = \text{span}\{\text{columns of } A\}$; $N(\mathbf{A}) = \{\mathbf{x} : \mathbf{Ax} = 0\}$.

**A17.** $n = \text{rank}(\mathbf{A}) + \dim N(\mathbf{A})$ (equivalently $n = \text{rank}(\mathbf{A}) + \text{nullity}(\mathbf{A})$).

**A18.** $ad - bc$; absolute value = area scaling factor for the unit square (signed area accounts for orientation).

**A19.** **False** — $\det \mathbf{A} = 0$ means noninvertible.

**A20.** Independent spanning list; dimension = number of vectors in any basis.

**A21.** (a) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$

**A22.** $\mathbf{E}_1$: multiply row 2 by 5. $\mathbf{E}_2$: swap the two rows. $\mathbf{E}_3$: add 4 times row 2 to row 1.

**A23.** $\mathbf{A}^{-1} = \dfrac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$; division by zero / no inverse when $\det \mathbf{A} = 0$.

**A24.** (a) $m$-vector (b) $m \times n$ matrix (c) **undefined** — inner dimensions $3 \neq 2$ (d) $n \times m$ matrix (e) **scalar** (f) **undefined** — singular, no inverse (g) **set of vectors** (h) **set of vectors** (i) **scalar** (nonnegative integer) (j) **undefined** — $\mathbf{AA}$ sizes fail (k) 2-vector (l) $m$-vector that is the zero vector when consistent (m) $2 \times 3$ matrix (n) **scalar** equal to 3 ($n - \text{rank} = 5 - 2$)

**A25.** **False** — $N(\mathbf{A}) = \{0\}$ gives uniqueness when a solution exists, but $\mathbf{b}$ might miss $C(\mathbf{A})$ (e.g. tall full-column-rank $\mathbf{A}$).

**A26.** 1–a, 2–b, 3–a, 4–a, 5–a.

**A27.** Fraction (or amount) moving from state $j$ into state $i$ in one step.

## B — Easy computations

**B1.** $\mathbf{Ax} = 2\begin{bmatrix}1\\0\end{bmatrix} + 1\begin{bmatrix}-1\\3\end{bmatrix} + (-1)\begin{bmatrix}2\\1\end{bmatrix} = \begin{bmatrix}-1\\2\end{bmatrix}$.

**B2.** $\mathbf{Ax} = \begin{bmatrix}4\\-5\end{bmatrix}$.

**B3.** $\mathbf{Ax} = \begin{bmatrix}1\\3\\-2\end{bmatrix}$.

**B4.** Input $\mathbf{x}$ is $3 \times 1$ (in $\mathbb{R}^3$); output $\mathbf{y}$ is $2 \times 1$ (in $\mathbb{R}^2$).

**B5.** $\mathbf{y}_1 = (0, 1)$, $\mathbf{y}_2 = (-1, 0)$; sketch shows basis vectors rotated 90° counterclockwise. Effect of $\mathbf{y} = \mathbf{Ax}$: 90° counterclockwise rotation.

**B6.** $\mathbf{y}_1 = (1, 0)$, $\mathbf{y}_2 = (2, 1)$; sketch shows $\mathbf{x}_1$ fixed and $\mathbf{x}_2$ sent to $(2, 1)$. Effect of $\mathbf{y} = \mathbf{Ax}$: **shear** (horizontal shear).

**B7.** $\mathbf{A} + \mathbf{B} = \begin{bmatrix}2 & 1\\1 & 3\end{bmatrix}$; $2\mathbf{A} = \begin{bmatrix}4 & 0\\0 & 6\end{bmatrix}$; $\mathbf{A}^\mathsf{T} = \mathbf{A}$.

**B8.** Symmetric: yes. Triangular: no. Diagonal: no.

**B9.** $\mathbf{DF}\begin{bmatrix}1\\2\end{bmatrix} = \begin{bmatrix}-2\\1\end{bmatrix}$.

**B10.** $\mathbf{A}^\mathsf{T} = \begin{bmatrix}1 & 3\\0 & 1\\2 & 0\end{bmatrix}$, size $3 \times 2$.

**B11.** $\mathbf{RP} = \begin{bmatrix}0 & 0 & -1\\0 & 1 & 0\end{bmatrix}$.

**B12.** $\mathbf{A} = \begin{bmatrix}1 & 2\\3 & -1\end{bmatrix}$, $\mathbf{x} = \begin{bmatrix}x\\y\end{bmatrix}$, $\mathbf{b} = \begin{bmatrix}5\\1\end{bmatrix}$.

**B13.** Unique solution: $\mathbf{x} = 1$, $y = 2$.

**B14.** Pivots $\mathbf{x}_1, \mathbf{x}_2$; free $\mathbf{x}_3 = t$. General: $\mathbf{x} = (3 - 2t,\ 1 + t,\ t)$.

**B15.** **No solutions** — inconsistent $(0 = 4)$.

**B16.** RREF $\left[\begin{array}{cc|c}1 & 2 & 3\\0 & 0 & 0\end{array}\right]$; infinitely many: $\mathbf{x}_1 = 3 - 2t$, $\mathbf{x}_2 = t$.

**B17.** **No** — second row of $\mathbf{A}$ is $2\times$ first, but $7 \neq 2 \cdot 3$.

**B18.** Pivot vars $\mathbf{x}_1, \mathbf{x}_3$; free $\mathbf{x}_2 = s$, $\mathbf{x}_4 = t$; rank 2; yes consistent;

$$\mathbf{x} = \begin{bmatrix}2\\0\\-1\\0\end{bmatrix} + s\begin{bmatrix}-3\\1\\0\\0\end{bmatrix} + t\begin{bmatrix}1\\0\\-4\\1\end{bmatrix}.$$

**B19.** Result $\left[\begin{array}{cc|c}1 & 2 & 3\\0 & 0 & 0\end{array}\right]$; infinitely many: $\mathbf{x}_1 = 3 - 2t$, $\mathbf{x}_2 = t$.

**B20.** $\mathbf{x}_1 = (0.5, 0.5)$; $\mathbf{x}_2 = (0.5, 0.5)$.

**B21.** $\mathbf{x}_1 = (0.62, 0.38)$; urban / rural fractions tomorrow.

**B22.** $\mathbf{x}_3 = \mathbf{A}^3\mathbf{x}_0$.

**B23.** (a) $\mathbf{A} = \begin{bmatrix}0.7 & 0.4\\0.3 & 0.6\end{bmatrix}$. (b) $\mathbf{x}_1 = (55, 45)$. (c) Yes — total 100.

**B24.** $\mathbf{AB} = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$, $\mathbf{BA} = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix}$; **not** equal.

**B25.** $\mathbf{A}^2 = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$.

**B26.** $\mathbf{A}^2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, $\mathbf{A}^3 = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$; $\mathbf{A}^k = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$.

**B27.** $\mathbf{A}^2 = \begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix}$, $\mathbf{A}^3 = \begin{bmatrix} 8 & 0 \\ 0 & 27 \end{bmatrix}$; $\mathbf{A}^k = \begin{bmatrix} 2^k & 0 \\ 0 & 3^k \end{bmatrix}$.

**B28.** $\mathbf{A}^2 = 0$, so $\mathbf{A}^3\mathbf{x} = 0$; after two steps the state is already 0 (same as $\mathbf{x}_2 = \mathbf{A}^2\mathbf{x}_0$).

**B29.** $[\mathbf{A} \mid \mathbf{I}] = \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{array}\right] \to \left[\begin{array}{cc|cc} 1 & 0 & 1 & -2 \\ 0 & 1 & 0 & 1 \end{array}\right]$, so $\mathbf{A}^{-1} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$.

**B30.** **Not invertible**: $\det = 0$; nontrivial null space, e.g. $(2, -1)$.

**B31.** $\mathbf{x}_2 = 2$, $\mathbf{x}_1 = 1$.

**B32.** $\det = 1$; $\mathbf{A}^{-1} = \begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix}$; product is $\mathbf{I}$.

**B33.** $\mathbf{A}^{-1} = \begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix}$; $\det \mathbf{A} = 1$.

**B34.** $C(\mathbf{A}) = \text{span}\{(1,2,0),(0,0,1)\} = \{(\mathbf{x}, 2\mathbf{x}, z)\}$. $(1,2,3)$: **yes**; $(1,3,0)$: **no**.

**B35.** $N(\mathbf{A}) = \{0\}$ (two pivot columns).

**B36.** Rank 2; pivot columns 1 and 3. Free $\mathbf{x}_2 = s$, $\mathbf{x}_4 = t$:

$$\mathbf{x} = s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ 1 \\ 1 \end{bmatrix}.$$

**B37.** $\dim N(\mathbf{A}) = 2$; $2 + 2 = 4 = n$.

**B38.** (a) Rank 1. (b) $C(\mathbf{A}) = \text{span}\{(1,2)\}$; $N(\mathbf{A}) = \text{span}\{(-2,1,0),(-3,0,1)\}$. (c) $(1,2)$ yes; $(1,3)$ no.

**B39.** $12 - 2 = 10$.

**B40.** Product of diagonal: $2 \cdot (-3) \cdot 5 = -30$.

**B41.** Area 6; equals $|\det \mathbf{A}| = 6$.

**B42.** Unique: $\mathbf{x} = 2$, $y = 1$.

**B43.** Free $\mathbf{x}_2 = t$; $\mathbf{x} = (4 - 2t,\ t,\ -1)$; $\text{rank}(\mathbf{A}) = 2$.

**B44.** (a) Columns span $\mathbb{R}^2$; e.g. basis $\{(1,0),(0,1)\}$ or the first two columns; rank 2. (b) $N(\mathbf{A}) = \text{span}\{(-1,1,-1)\}$ (equivalently $\text{span}\{(1,-1,1)\}$). (c) $2 + 1 = 3 = n$.

**B45.** $\det = 1 \cdot 5 - 2 \cdot 3 = -1$; $\mathbf{A}^{-1} = \begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}$.

## C — Word problems

**C1.** (a) $x + y = 120$, $4x + 3y = 430$. (b) $\left[\begin{array}{cc|c} 1 & 1 & 120 \\ 4 & 3 & 430 \end{array}\right]$. (c) e.g. $R_2 \leftarrow R_2 - 4R_1$ gives $\left[\begin{array}{cc|c} 1 & 1 & 120 \\ 0 & -1 & -50 \end{array}\right]$ (echelon). (d) $y = 50$, $x = 70$. (e) **70** lattes and **50** drip coffees.

**C2.** (a) $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$, $\mathbf{x} = (x_1, x_2)$, $\mathbf{b} = (16, 13)$. (b) Subtract $\frac{1}{2}$ times row 1 from row 2: $\mathbf{E} = \begin{bmatrix} 1 & 0 \\ -1/2 & 1 \end{bmatrix}$, $\mathbf{U} = \begin{bmatrix} 2 & 1 \\ 0 & 5/2 \end{bmatrix}$. (c) $\mathbf{L} = \mathbf{E}^{-1} = \begin{bmatrix} 1 & 0 \\ 1/2 & 1 \end{bmatrix}$; $\mathbf{LU} = \mathbf{A}$. (d) $\mathbf{Ly} = \mathbf{b}$ gives $\mathbf{y} = (16, 5)$; $\mathbf{Ux} = \mathbf{y}$ gives $x_2 = 2$, $x_1 = 7$. (e) **7** orders of A and **2** orders of B.

**C3.** (a) $\mathbf{A} = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}$. (b) $\mathbf{x}_1 = (190, 110)$. (c) Yes — total 300 (each column of $\mathbf{A}$ sums to 1, so totals are preserved). (d) $\mathbf{x}_2 = \mathbf{A}^2\mathbf{x}_0$.

**C4.** (a) $C(\mathbf{A}) = \text{span}\{(1, 2)\}$ (second column = 2× first); $N(\mathbf{A}) = \text{span}\{(2, -1)\}$ (solve $\mathbf{Ax} = 0$). (b) $C(\mathbf{A})$: the line through the origin in direction $(1, 2)$. $N(\mathbf{A})$: the line through the origin in direction $(2, -1)$. (c) $\text{rank}(\mathbf{A}) = 1$, $\dim N(\mathbf{A}) = 1$, and $1 + 1 = 2 = n$. (d) Possible outputs $\mathbf{y}$ lie only on that line in the (parts, scrap)-plane; when $\mathbf{y}$ is achievable, $\mathbf{x}$ is not unique (you can add any multiple of $(2, -1)$).